

CERTIFIED RANK-DEPTH SELECTION FOR BLOCK MATRIX-FUNCTION GRADIENTS

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Abstract. Inverse problems with correlated preparations require block matrix-function responses and parameter derivatives. For cosine responses of symmetric affine operators, we derive a loss-aware compression bound whose retained-block contribution cancels, giving quadratic dependence on the discarded probe norm for consistent targets. Separate response and operator-derivative Chebyshev remainders bound the inexact loss gradient before operator applications. A finite search minimizes nominal Hamiltonian-vector work mr over certified rank-depth pairs. On four authenticated electronic-structure Hamiltonians of orders 769–19,896, the refinements reduce joint certified work from 120 to 48 vector applications, a 60 percent saving; independent gradient discrepancies are below 9×10^{-9} at tolerance 10^{-3} . Certification nevertheless costs more than an incremental full-block stopping heuristic at that tolerance. A larger simulated inverse study crosses two chain sizes, three truths, two starts, two noisy realizations plus clean data, two stationarity targets, and four methods. All 288 runs recover parameters within 10^{-3} , with active compression during certified feasible-descent iterations. Strict final stationarity can require full probe rank and remove the work benefit. The original 96-site recovery failures are retained and diagnosed independently. The results establish a sharper exact-arithmetic resource guarantee and reduced learning computations, while separating these benefits from runtime, finite-precision certification, and global identifiability.

1. Introduction. Matrix-function inverse problems arise in quantum dynamics, response calculations, and scientific computing [2–4]. For a large symmetric operator $A(\theta)$ and a thin preparation block B , the observations may involve every entry of $B^\top f(A(\theta))B$. Correlated columns create an opportunity to reduce the width of each operator application. The difficulty is to account for the resulting perturbation of the loss gradient while also choosing a sufficient Krylov depth.

Block Krylov projection and its matrix-function error analysis are established [7, 8, 16]. Fréchet derivatives admit augmented, low-rank, and bivariate Krylov formulations [9–11]. Singh, Barros, and Li [12] develop a forward-only gradient approximation for Hermitian bilinear forms, including learning examples. Scalar bilinear responses can also be reconstructed by polarization, so block output alone does not define the novelty of the present method.

Our contribution is a resource-selection guarantee for cosine block gradients that combines a loss-aware compression bound with separate response and operator-derivative Chebyshev tails. The supporting norm inequalities are elementary; their integration permits a finite search over rank and depth without extending the large Krylov space. We distinguish response exactness through degree $2m - 1$ from derivative exactness through m , and minimize mr over feasible pairs. This is a grid optimum for the specified work model, not a minimum-runtime theorem.

The experiments answer three separate questions: whether compression reduces work at the same depth; whether it reduces work at the same certified tolerance; and whether certification is competitive with methods that achieve the same observed accuracy. A simulated lattice inverse problem then tests whether the inexact gradients support parameter recovery. Authentic Hamiltonian inputs, constructed numerical probes, and simulated physical observations are explicitly distinguished. No quantum computational advantage is claimed.

2. Objective and access model. Let $A(\theta) = H_0 + \sum_{j=1}^p \theta_j D_j$ be real symmetric, with $|\theta_j| \leq a_j$, and let $B \in \mathbb{R}^{n \times b}$. Define

$$F_\ell(\theta) = B^\top \cos(t_\ell A(\theta))B, \quad \Phi(\theta) = \frac{1}{2s} \sum_{\ell=1}^s \|F_\ell - Y_\ell\|_F^2. \quad (2.1)$$

The exact gradient is

$$g_j = \frac{1}{s} \sum_{\ell} \langle F_\ell - Y_\ell, B^\top \mathcal{L}_{f_\ell}(A, D_j)B \rangle_F, \quad f_\ell(x) = \cos(t_\ell x). \quad (2.2)$$

Here $\mathcal{L}_f$ is the Fréchet derivative [2, 13]. The implementation needs block actions of A and of each D_j . Diagonal fields and symmetric sparse off-diagonal directions are supported.

Lemma 1 (Uniform enclosure). *If $\|H_0\|_\infty + \sum_j a_j \|D_j\|_2 \leq \rho$, then $\|A(\theta)\|_2 \leq \rho$ on the entire parameter box.*

Proof. Symmetry implies $\|H_0\|_2 \leq \|H_0\|_\infty$. Apply the triangle inequality and $|\theta_j| \leq a_j$. A diagonal direction has spectral norm $\max_i |(D_j)_{ii}|$; the maximum absolute row sum bounds a symmetric sparse direction's spectral norm. $\square$

No eigenvalue iteration is required. This is an exact-arithmetic statement; floating-point row sums are not interval enclosures. The cosine analysis does not require $\rho < 1$: the approximation difficulty depends on $|t|\rho$. Rescaling a physical Hamiltonian by c must be accompanied by rescaling time by $1/c$ to preserve the evolution. The PARSEC times below are dimensionless benchmark inputs, not experimental times.

3. Compression, projection, and certification.

3.1. Probe compression. Write the thin SVD as $B = U\Sigma W^\top$ and its rank- r truncation as $B_r = U_r \Sigma_r W_r^\top = C_r W_r^\top$. Let $\delta_r = \|B - B_r\|_F$ and $\alpha_r = 2\|B\|_2 \delta_r$.

Theorem 1 (Compression perturbation). *For real symmetric A, D ,*

$$\|B^\top \cos(tA)B - B_r^\top \cos(tA)B_r\|_F \leq \alpha_r, \quad (3.1)$$

$$\|B^\top \mathcal{L}_{f_t}(A, D)B - B_r^\top \mathcal{L}_{f_t}(A, D)B_r\|_F \leq |t| \|D\|_2 \alpha_r. \quad (3.2)$$

Proof. Put $E = B - B_r$. For any M , $B^\top MB - B_r^\top MB_r = E^\top MB + B_r^\top ME$. Frobenius-spectral product inequalities bound its norm by $(\|B\|_2 + \|B_r\|_2) \|M\|_2 \delta_r = \alpha_r \|M\|_2$. For $M = \cos(tA)$ the norm is at most one. Differentiating $e^{it(A+sD)}$ by the Duhamel formula and using unitary invariance gives $\|\mathcal{L}_{e^{it(\cdot)}}(A, D)\|_2 \leq |t| \|D\|_2$. Averaging the two exponential derivatives proves the cosine bound. $\square$

Set $c_B = \|B\|_2 \|B\|_F$ and $c_r = \|B_r\|_2 \|B_r\|_F$. The exact and compressed exact gradients differ componentwise by at most

$$\chi_{j,r} = \frac{\alpha_r \|D_j\|_2}{s} \sum_\ell |t_\ell| (c_B + c_r + \|Y_\ell\|_F). \quad (3.3)$$

This follows by adding and subtracting the compressed response times the full derivative in Eq. (2.2), then applying the preceding theorem.

The loss permits a sharper estimate because its retained-retained gradient contribution cancels exactly. Complete W_r to an orthogonal matrix $[W_r \ W_d]$, put $C_d = BW_d$, and define $u_r = \|C_r\|_2 \|C_d\|_F$ and $v_r = \|C_d\|_2 \|C_d\|_F$. Zero singular components may be included in C_d when $b > n$. Assume the targets are symmetric, as are the observations. For nonsymmetric targets, their symmetric parts give the same gradient.

Theorem 2 (Loss-aware compression). *Writing $Y_{\ell,rd} = W_r^\top Y_\ell W_d$ and $Y_{\ell,dd} = W_d^\top Y_\ell W_d$, one has*

$$|g_j - g_j^{(r)}| \leq \tilde{\chi}_{j,r} := \frac{\|D_j\|_2}{s} \sum_\ell |t_\ell| [2u_r(u_r + \|Y_{\ell,rd}\|_F) + v_r(v_r + \|Y_{\ell,dd}\|_F)]. \quad (3.4)$$

For consistent noiseless targets $Y_\ell = B^\top \cos(t_\ell A_\star)B$, the bracket is at most $4u_r^2 + 2v_r^2$.

Proof. Transform both factors of the Frobenius inner product in Eq. (2.2) by $[W_r \ W_d]$. The retained-retained terms are identical for B and B_r ; the difference consists of twice the retained-discarded term and the discarded-discarded term. Their response norms are at most u_r, v_r , and their derivative norms at most $|t_\ell| \|D_j\|_2 u_r, |t_\ell| \|D_j\|_2 v_r$. Cauchy-Schwarz gives Eq. (3.4). Consistent target blocks obey the same response bounds. Thus the consistent-target bound is quadratic in the discarded probe norm; arbitrary noise is handled by the actual target blocks. $\square$

The default compression bound is $\bar{\chi}_{j,r} = \min\{\chi_{j,r}, \tilde{\chi}_{j,r}\}$ componentwise. The linear bound is retained as an ablation. Target block norms are evaluated with $P_d = I - W_r W_r^\top$, avoiding an additional orthogonal completion.

3.2. Distinct response and derivative exactness. Let $m \geq 2$ and define $\mathcal{K}_m(A, C_r) = \text{span}\{C_r, AC_r, \dots, A^{m-1}C_r\}$. Let V be an orthonormal basis of this space, and write $T = V^\top AV$, $C = V^\top C_r$, and $Q = CW_r^\top$. The lifted approximations are

$$\hat{F}_\ell = Q^\top \cos(t_\ell T)Q, \quad \hat{J}_{\ell j} = Q^\top \mathcal{L}_{f_\ell}(T, V^\top D_j V)Q, \quad \hat{g}_j = \frac{1}{s} \sum_\ell \langle \hat{F}_\ell - Y_\ell, \hat{J}_{\ell j} \rangle_F. \quad (3.5)$$

We call $\hat{g}$ an *inexact gradient of the original loss*. It generally is not the derivative of the computed projected loss when rebuilding V as θ changes; that derivative also contains basis-variation terms.

Theorem 3 (Polynomial exactness). *The identity $C_r^\top q(A)C_r = C^\top q(T)C$ holds for every polynomial of degree at most $2m - 1$. The identity $C_r^\top \mathcal{L}_q(A, D)C_r = C^\top \mathcal{L}_q(T, V^\top DV)C$ holds for degree at most m .*

Proof. For $0 \leq k \leq m - 1$, $A^k C_r = VT^k C$. For a response monomial of degree at most $2m - 2$, split the degree into two powers at most $m - 1$ and use symmetry. For degree $2m - 1$, use $(A^{m-1}C_r)^\top A(A^{m-1}C_r) = C^\top T^{2m-1}C$. For a derivative monomial of degree $k \leq m$, $\mathcal{L}_{x^k}(A, D) = \sum_{u=0}^{k-1} A^u D A^{k-1-u}$; both powers are at most $m - 1$, so substitution proves the identity term by term. Linearity completes the proof. The derivative assertion does not in general extend to $2m - 1$. $\square$

Corollary 1 (Invariant-space termination). *If $AV = VT$ and C_r lies in the range of V , both compressed responses and sandwiched derivatives are exact for cosine. In particular this holds when V is square and orthogonal.*

Proof. Invariance gives $A^k C_r = VT^k C$ for every k . Apply the monomial derivative identity termwise to the absolutely convergent cosine series. $\square$

This statement concerns exact invariance. A small numerical factorization residual is not by itself a proof of zero error; the implementation removes the projection tail only for a square computed basis, under its stated exact-arithmetic interpretation.

3.3. Gradient certificate. For $\rho > 0$, define the positive Taylor tails

$$R_d^{(0)}(t, \rho) = \sum_{\substack{\nu > d \\ \nu \text{ even}}} \frac{|t|^\nu \rho^\nu}{\nu!}, \quad R_d^{(1)}(t, \rho) = \sum_{\substack{\nu > d \\ \nu \text{ even}}} \frac{\nu |t|^\nu \rho^{\nu-1}}{\nu!}. \quad (3.6)$$

The $\rho = 0$ cosine remainders vanish. To reduce the required depth, also use the Chebyshev expansion

$$\cos(zx) = J_0(z) + 2 \sum_{k \geq 1} (-1)^k J_{2k}(z) T_{2k}(x), \quad z = |t|\rho, \quad x \in [-1, 1]. \quad (3.7)$$

For a real argument, $|J_\nu(z)| \leq (z/2)^\nu / \nu!$ [1]. Therefore define the alternative majorants

$$E_d^{(0)} = 2 \sum_{\substack{\nu > d \\ \nu \text{ even}}} \frac{(z/2)^\nu}{\nu!}, \quad E_d^{(1)} = \frac{2}{\rho} \sum_{\substack{\nu > d \\ \nu \text{ even}}} \frac{\nu^2 (z/2)^\nu}{\nu!}. \quad (3.8)$$

The derivative estimate is an operator bound, not merely a scalar derivative bound. Differentiating the Chebyshev recurrence for a symmetric contraction X gives

$$\mathcal{L}_{T_\nu}(X, D) = U_{\nu-1}(X)D + 2 \sum_{k=1}^{\nu-1} U_{\nu-1-k}(X)DT_k(X). \quad (3.9)$$

Here U_k is the Chebyshev polynomial of the second kind. Since $\|T_k(X)\|_2 \leq 1$ and $\|U_k(X)\|_2 \leq k + 1$, its norm is at most $\nu^2 \|D\|_2$. Scaling $X = A/\rho$ proves the derivative remainder bound in Eq. (3.8). These tails use only the same uniform enclosure; they require no spectral estimation. In the following theorem $R^{(0)}, R^{(1)}$ may be replaced by $E^{(0)}, E^{(1)}$, respectively.

Theorem 4 (Preflight gradient bound). *Under the uniform enclosure, the inexact gradient in Eq. (3.5) satisfies, in exact arithmetic,*

$$\|g - \hat{g}\|_2 \leq \|\eta_{m,r}\|_2, \quad \eta_{j,m,r} = \bar{\chi}_{j,r} + \kappa_{j,m,r}, \quad (3.10)$$

$$\begin{aligned} \kappa_{j,m,r} = \frac{1}{s} \sum_{\ell} [2c_r R_{2m-1}^{(0)}(t_\ell, \rho) c_r |t_\ell| \|D_j\|_2 \\ + (c_r + \|Y_\ell\|_F) 2c_r \|D_j\|_2 R_m^{(1)}(t_\ell, \rho)]. \end{aligned} \quad (3.11)$$

All bound inputs are available before an application of A .

Proof. Subtract the Taylor polynomials of degrees $2m - 1$ and m from the response and derivative, respectively. Polynomial exactness cancels their contributions. Since $\|T\|_2 \leq \rho$, the two full-minus-projected remainders are bounded by $2c_r R_{2m-1}^{(0)}$ and $2c_r \|D_j\|_2 R_m^{(1)}$. The compressed exact derivative norm is at most $c_r |t_\ell| \|D_j\|_2$, and the projected residual norm is at most $c_r + \|Y_\ell\|_F$. Adding and subtracting the compressed exact gradient yields κ by Cauchy–Schwarz; either compression bound supplies χ . Combine the component bounds in Euclidean norm. $\square$

Chebyshev remainders follow by subtracting their corresponding polynomials and applying the same polynomial exactness argument. The revised default uses loss-aware compression and Chebyshev tails. Four combinations of the two compression bounds and two tail bounds isolate their effects. The evaluated-residual variant replaces $c_r + \|Y_\ell\|_F$ by $\|\hat{F}_\ell - Y_\ell\|_F$, but requires the projected calculation. The preflight variant trades tighter stopping for avoiding those repeated costs.

3.4. Resource selection and optimization. For a finite increasing grid $\mathcal{M} \subset \{2, 3, \dots\}$, choose

$$(r_\star, m_\star) \in \arg \min_{1 \leq r \leq \min(n,b), m \in \mathcal{M}} \{mr : \|\eta_{m,r}\|_2 \leq \varepsilon\}, \quad (3.12)$$

where only ranks with nonzero retained singular values are executable; ties favor smaller r and then smaller m . Only the first feasible depth at each rank need be compared because the work increases with depth. Empty feasible sets raise an explicit failure; no uncanceled answer is silently accepted. We also evaluate a full-rank restriction and a rank-first policy allocating fractions 0.05, 0.2, or 0.5 of the tolerance to compression. Equation (3.12) minimizes the grid work, not all possible Krylov algorithms or measured CPU time.

With fixed probes, targets, times, directions, parameter box, and tolerance, the plan is independent of θ . The learning experiment caches it within each tolerance stage and recomputes it only when the tolerance changes.

Corollary 2 (Descent condition). *If $\|g - \hat{g}\|_2 \leq \varepsilon$ and $\|\hat{g}\|_2 > \varepsilon$, then $-\hat{g}$ is a strict descent direction for the original differentiable loss.*

Proof. $g^\top \hat{g} = \|\hat{g}\|_2^2 + (g - \hat{g})^\top \hat{g} \geq \|\hat{g}\|_2 (\|\hat{g}\|_2 - \varepsilon) > 0$. $\square$

This motivates tightening tolerances near stationarity. It does not establish global convergence of the optimizer or parameter identifiability. For a general feasible trial direction d , the same certificate yields

$$g^\top d \leq \hat{g}^\top d + \varepsilon \|d\|_2. \quad (3.13)$$

Thus a negative right-hand side certifies descent even when $d \neq -\hat{g}$. The larger learning experiment tightens the tolerance until this test passes. This direction test alone does not certify line-search acceptance; the independent floating-point loss computations used for acceptance and their costs are reported separately.

3.5. Implementation and finite precision. The basis is fully reorthogonalized. Applied blocks are cached and reused to form T , so the operator ledger is mr in the absence of deflation, or the sum of actual block widths otherwise. The basis occupies $O(nmr)$ entries; this count is not a measurement of peak process memory. Rank loss is detected by the QR threshold 10^{-12} ; early breakdown stops basis growth. The floating-point threshold and basis perturbations are outside the exact-arithmetic theorem, so orthogonality defects are reported separately.

One eigendecomposition $T = SAS^\top$ is reused across all times and directions. Divided differences use

$$\frac{\cos(tx) - \cos(ty)}{x - y} = -t \sin\left(\frac{t(x+y)}{2}\right) \operatorname{sinc}\left(\frac{t(x-y)}{2}\right), \quad \operatorname{sinc}(z) = \frac{\sin z}{z}, \quad (3.14)$$

with the continuous value at zero. Directions are projected and rotated once; the $b \times b$ responses and derivatives are then contracted in the loss. Writing $k \leq mr$ for basis size, the principal costs are operator actions, $O(nk^2)$ reorthogonalization, $O(pnk^2)$ projection for diagonal directions, one $O(k^3)$ eigendecomposition, and $O(pk^3)$ direction rotations. Subsequent responses and derivatives use small matrix contractions rather than repeating eigendecompositions sp times. General sparse directions additionally incur their block-action costs. Both full and compressed baselines use this implementation.

4. Experimental protocol.

4.1. Authenticated operator benchmark. The four PARSEC Hamiltonians come from the SuiteSparse Matrix Collection [14, 15]. Downloads are authenticated by pinned SHA-256 archive digests before parsing. The development/confirmation labels identify the original split, not prospective preregistration of this revision. The Hamiltonians are scaled to row-sum norm 0.8. Three diagonal Gaussian index

Table 1: Authenticated inputs and uniform parameter-box bounds.

matrix	n	nnz	role	ρ
Si2	769	17,801	development	0.979996
SiH4	5,041	171,903	confirmation	0.980000
benzene	8,219	242,669	confirmation	0.980000
Si5H12	19,896	738,598	confirmation	0.980000

fields have maximum magnitude at most 0.12, with $|\theta_j| \leq 0.5$. Eight normalized overlapping Gaussian index envelopes form B . These fields and preparations are reproducible numerical constructions, not atomic-coordinate models. The target parameter is $(0.32, -0.24, 0.16)$ and gradients are evaluated at $(0.06, -0.03, 0.02)$ for times $(2, 4, 6)$.

Targets and reference gradients are computed independently by sparse augmented exponential actions (Appendix B). The primary certified tolerance is 10^{-3} with depths $\{4, 6, \dots, 28\}$. Observed-accuracy frontiers use $\{2, 4, 6, 8, 12, 16, 24, 28\}$. Scalar polarization uses $\{2, 4, 8, 12\}$, a disclosed smaller grid. The time-tolerance map uses depths $\{4, 6, \dots, 42\}$ and horizons 2, 4, 6, 8; these are separate from the primary grid.

4.2. Comparators and timing. We compare joint selection, certified full rank, three rank-first budgets, and full/selected-rank depth frontiers. A scalar forward-only projected Fréchet baseline assembles diagonal and off-diagonal responses by

$$b_i^\top f(A) b_j = \frac{1}{2} [(b_i + b_j)^\top f(A) (b_i + b_j) - b_i^\top f(A) b_i - b_j^\top f(A) b_j]. \quad (4.1)$$

It uses $b(b+1)/2$ scalar factorizations. This is a reorthogonalized polarization comparator, not a reproduction of the unreorthogonalized implementation in Ref. [12]. An incremental successive-depth heuristic extends one nested basis and stops after two consecutive gradient changes are at most $\varepsilon/4$. Every checkpoint assembly is timed; applied operator blocks are counted once. A full-rank version and a version with the rank frozen at its primary 10^{-3} choice are recorded at all three accuracy levels. The latter cannot control a fixed compression bias at a tighter tolerance. Neither has a rigorous stopping guarantee. Central finite differences with step 10^{-5} remain an auxiliary check.

We also implement a projected-adjoint contraction. In the eigenbasis of T , the time-aggregated Loewner sensitivity is formed once; each gradient component is then obtained by contracting its lifted sensitivity with $D_j V$. This avoids separate direction projections and rotations. It computes the same fixed-projector estimator, as checked against explicit derivatives, and is our implementation rather than a replication of published software.

The post hoc oracle chooses the least recorded work meeting each observed error threshold $10^{-3}, 10^{-6}, 10^{-9}$. It requires the independent reference and is unavailable during normal execution. By contrast, the certified comparison uses only Eq. (3.11). These are different performance questions.

The replay uses Python 3.12.13, NumPy 2.5.3, and SciPy 1.18.1 on an Apple M4 Pro running macOS 26.6.2 with one requested BLAS thread; backend metadata are retained. Joint versus same-depth full rank uses nine paired repetitions with alternating order. Other timings use three repetitions in fixed method order. Timings include SVD, selection where applicable, basis construction, audits, and reduced assembly, and exclude data loading and reference generation. The independent reference runtime is reported separately. Ranges show observed repetition variability, not confidence intervals across machines.

5. Computational results.

Table 2: Joint certified selection at 10^{-3} . Error is discrepancy from the independent augmented-exponential reference. Ratio is the median same-depth full-block time divided by joint-selection time.

matrix	n	rank	depth	bound	error	work	ratio
Si2	769	3	16	5.11e-04	8.96e-09	48	0.95
SiH4	5,041	3	16	4.87e-04	1.55e-10	48	1.53
benzene	8,219	3	16	5.09e-04	3.98e-09	48	1.64
Si5H12	19,896	3	16	5.76e-04	2.87e-09	48	1.38

5.1. Certification versus achieved accuracy. Joint selection chooses ranks 3 and depths 16, using 48 operator-vector equivalents. This reduces work by 60.0 percent relative to the original linear/Taylor joint rule, and 62.5 percent relative to full-rank certification with the new tail bound. The same-depth median full/proposed timing ratios range from 0.95 to 1.64; selection overhead is included. The largest observed orthogonality defect is 8.01×10^{-15} . Table 3 separates the compression improvement from the depth improvement. The response and derivative columns are norms of their componentwise contributions to Eq. (3.11), rather than the unweighted scalar tails. The same-depth comparison isolates

Table 3: Certificate ablation at 10^{-3} . The three final columns separate compression, response, and derivative contributions. Each pair is selected independently on the same grid.

matrix	bounds	r	m	work	compression	response	derivative
Si2	linear/taylor	5	24	120	3.4e-04	3.6e-23	1.2e-04
Si2	loss-aware/taylor	3	24	72	3.5e-05	3.6e-23	1.2e-04
Si2	linear/chebyshev	5	16	80	3.4e-04	3.9e-19	4.8e-04
Si2	loss-aware/chebyshev	3	16	48	3.5e-05	3.9e-19	4.8e-04
SiH4	linear/taylor	5	24	120	7.8e-05	3.7e-23	1.2e-04
SiH4	loss-aware/taylor	3	24	72	7.6e-06	3.7e-23	1.2e-04
SiH4	linear/chebyshev	5	16	80	7.8e-05	3.9e-19	4.8e-04
SiH4	loss-aware/chebyshev	3	16	48	7.6e-06	3.9e-19	4.8e-04
benzene	linear/taylor	5	24	120	2.3e-04	3.7e-23	1.2e-04
benzene	loss-aware/taylor	3	24	72	2.0e-05	3.7e-23	1.2e-04
benzene	linear/chebyshev	5	16	80	2.3e-04	3.9e-19	4.9e-04
benzene	loss-aware/chebyshev	3	16	48	2.0e-05	3.9e-19	4.9e-04
Si5H12	linear/taylor	5	24	120	5.4e-04	3.6e-23	1.2e-04
Si5H12	loss-aware/taylor	3	24	72	9.0e-05	3.6e-23	1.2e-04
Si5H12	linear/chebyshev	5	18	90	5.4e-04	2.0e-23	1.4e-05
Si5H12	loss-aware/chebyshev	3	16	48	9.0e-05	3.9e-19	4.9e-04

compression; it does not establish an advantage at matched observed accuracy. In particular, the low-depth errors in Fig. 1 are already far below the uniform bound. Joint certification takes 4.0–8.6

Table 4: Matched observed and certified accuracy at 10^{-3} . The observed full-block choice is a post hoc minimum-work oracle on the tested grid. Time ratio compares joint certification to that oracle; values above one mean certification takes longer.

matrix	$\ g\ _2$	joint work	certified full work	oracle work	time ratio
Si2	1.35e-02	48	128	32	5.8
SiH4	1.17e-02	48	128	32	4.0
benzene	1.24e-02	48	128	16	7.1
Si5H12	3.26e-03	48	128	16	8.6

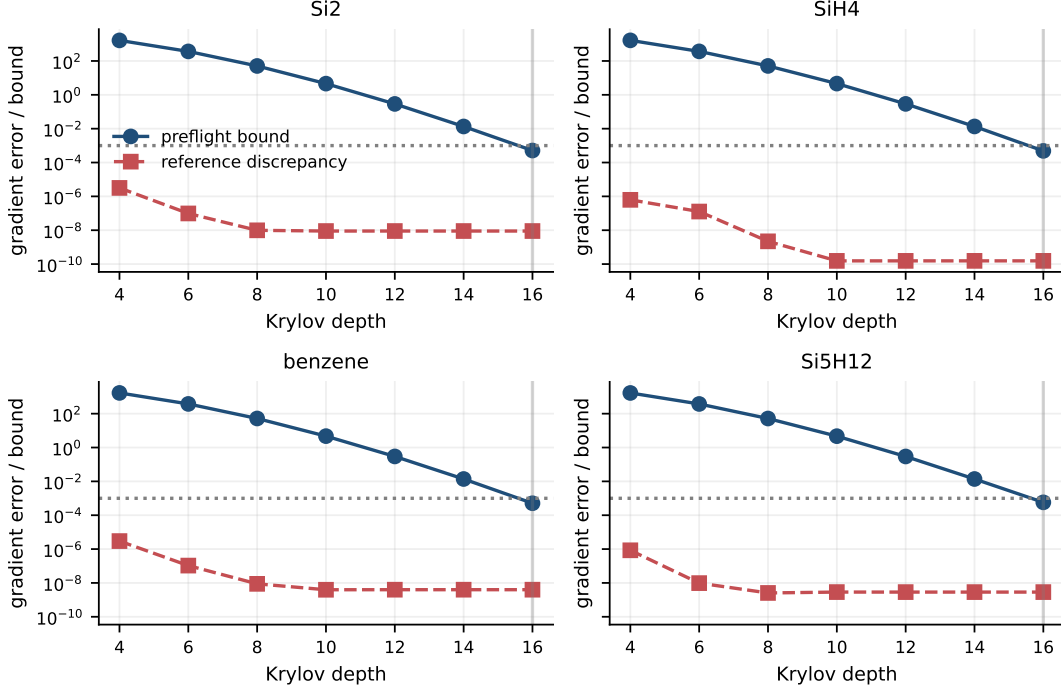


Fig. 1: Joint-plan rank held fixed while depth varies. Preflight bounds and independent-reference discrepancies are shown separately. The dotted line is the primary tolerance; the vertical line marks certified selection.

times as long as the full-block observed-accuracy oracle at 10^{-3} . Oracle search and reference costs are excluded, so this is a diagnostic frontier, not an executable stopping policy. Table 6 instead reports a basis-reusing executable comparator at three tolerances. At the primary tolerance the executable

Table 5: Rank-first budgets with the refined bounds. Entries are rank/depth/work; the budget fraction assigns that fraction of ε to compression.

matrix	0.05	0.2	0.5
Si2	3/16/48	3/16/48	3/16/48
SiH4	3/16/48	3/16/48	3/16/48
benzene	3/16/48	3/16/48	3/16/48
Si5H12	4/16/64	3/16/48	3/16/48

full-rank method uses 64 operator-vector applications, compared with 48 for joint certification, but joint certification takes 1.22–1.80 times as long. Its larger depth, selection costs, and different gradient assembly outweigh the width reduction in this comparison. The tighter-tolerance rows evaluate that stopping policy separately; the rank fixed at 10^{-3} is not a certified choice for those tighter targets. Absolute, relative, and normalized gradient-direction errors for each frontier point are retained in the result record. The full gradient norms in Table 4 make the absolute tolerance interpretable.

5.2. Resource sensitivity and unfavorable probes. The 36 stress cases use symmetric tridiagonal matrices of order 80; block widths 4, 8, 16; one or six symmetric off-diagonal directions; horizons 2 or 8; and probe singular values spanning 0, 1, or 3 decades. The tolerance is 10^{-5} . Independent augmented references are used in every case. All 36 discrepancies lie below their preflight bounds; the largest is 2.24×10^{-8} . 12 cases compress the probes and 17 reach the full dimension. The remaining cases and their selected dimensions are retained in the numerical records. These probes include orthogonal columns, so the study no longer tests only strongly compressible input. Full-space reductions and numerical deflation are counted by their actual applied widths. The result is a coverage and robustness study, not evidence of large-scale performance for general sparse directions.

Table 6: Executable incremental full-rank adjoint baseline. The stopping rule uses gradient agreement, while error is evaluated independently afterward. The last column is the post hoc full-adjoint oracle’s work, excluding its search and reference cost. Runtime includes every checkpoint calculation.

matrix	tolerance	depth	work	error	time (s)	oracle work
Si2	1e-03	8	64	1.6e-10	0.005	32
Si2	1e-06	12	96	3.0e-15	0.009	32
Si2	1e-09	16	128	3.2e-15	0.014	64
SiH4	1e-03	8	64	8.9e-12	0.023	32
SiH4	1e-06	8	64	8.9e-12	0.023	32
SiH4	1e-09	16	128	2.7e-15	0.062	48
benzene	1e-03	8	64	1.3e-10	0.035	16
benzene	1e-06	12	96	2.0e-14	0.060	32
benzene	1e-09	16	128	5.7e-16	0.094	64
Si5H12	1e-03	8	64	1.0e-11	0.087	16
Si5H12	1e-06	8	64	1.0e-11	0.086	32
Si5H12	1e-09	16	128	6.2e-16	0.233	48

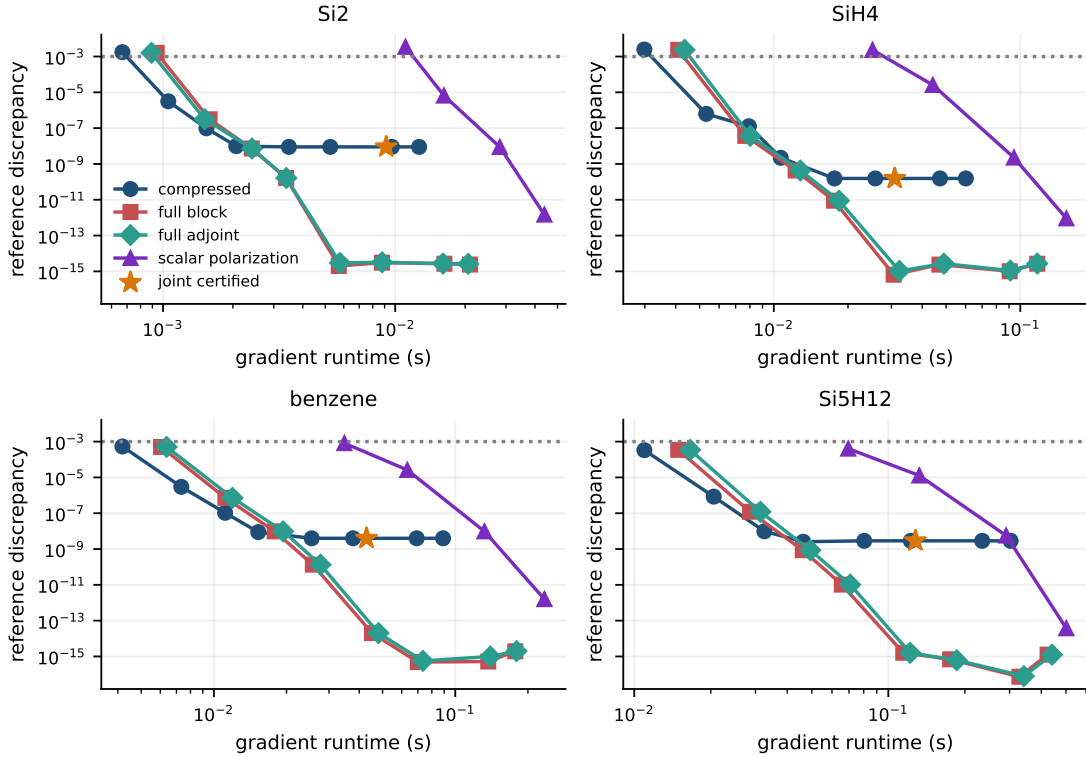


Fig. 2: Runtime versus discrepancy from the independent gradient. Curves show selected-rank blocks, full blocks, a full-block projected adjoint, and scalar polarization at explicitly sampled depths; stars show complete joint certification. Horizontal lines mark 10^{-3} . Points near the reference floor are numerical agreement, not proven absolute accuracy.

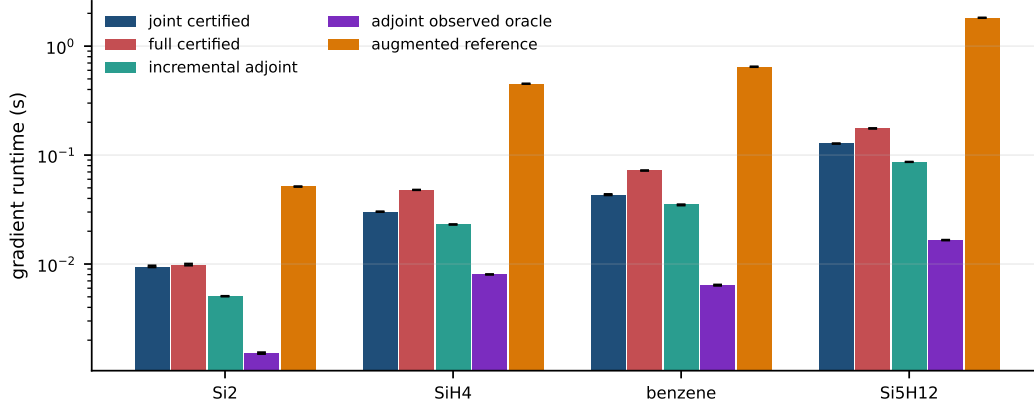


Fig. 3: Per-matrix runtimes: joint certification, certified full rank, incremental full adjoint, observed full-adjoint oracle at 10^{-3} , and independent augmented-exponential reference. Bars show medians and whiskers the minimum–maximum repetition range. The reference is an alternative numerical construction, not exact arithmetic.

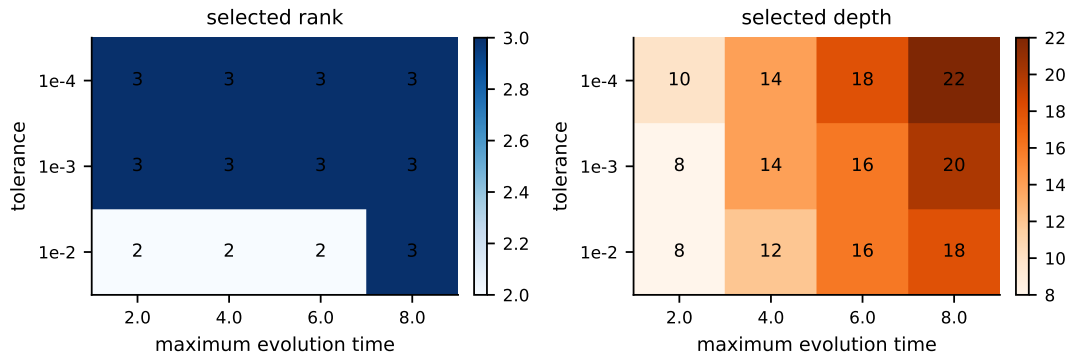


Fig. 4: Joint rank–depth choices for the Si2 time–tolerance grid, using the separate depth grid through 42. This is adaptation across inputs and tolerances, not dependence on the current parameter at fixed inputs.

5.3. Diagnostic recovery with the original certificate. Consider a single particle on a 96-site open chain, with nearest-neighbor adjacency K and Hamiltonian

$$A(\theta) = 0.35I - 0.22(1 + 0.3\theta_3)K + 0.25\theta_1 \text{diag}(v_{30}) + 0.25\theta_2 \text{diag}(v_{60}), \quad (v_c)_i = e^{-(i-c)^2/(2 \cdot 7^2)}. \quad (5.1)$$

Sites are indexed $0, \dots, 95$. Six normalized Gaussian wavepackets of width 6, centered at 24, 26, 28, 56, 58, 60, form the preparation block. Cosine block entries are real transition amplitudes of e^{-itA} for real preparations; access requires coherent amplitude/interference measurements rather than only transition probabilities. Here they are simulated without a device protocol. The noise model adds symmetric Gaussian amplitude perturbations with scale 10^{-4} ; it is not finite-shot or decoherence noise.

Two truths, $(0.22, -0.18, 0.12)$ and $(-0.16, 0.24, -0.14)$, are each fitted from three starts: zero, $(-0.3, 0.3, -0.25)$, and $(0.3, -0.25, 0.3)$. Training times are (2, 4, 6) and held-out times are (3, 5, 7). Joint and full-rank certification use cached plans at successive tolerances $10^{-3}, 10^{-5}, 10^{-7}$, with the original linear compression and Taylor bounds retained for this control. Each stage runs L-BFGS-B on $[-0.5, 0.5]^3$ with at most 80 iterations; a depth-4 full-block baseline uses the same optimizer schedule without certification. At a stage tolerance ε , its projected-gradient threshold is $\varepsilon/10$, function tolerance is 10^{-15} , and at most 30 line-search trials are allowed. These settings do not give an inexact-optimization convergence theorem. Optimizer status, every function-evaluation trajectory, work, and elapsed time are stored. Each recovery is timed once; variation across different starts is not a repeated-timing uncertainty estimate. Final losses and held-out predictions are recomputed independently.

Table 7: Simulated recovery: six runs per method and noise level (two truths, three starts). Parameter error and held-out relative response error are maxima; time is the median across those different runs. Recovery counts use parameter error below 10^{-3} .

method	noise	recovered	max. param. error	held-out error	time (s)
joint, original bound	0e+00	3/6	9.41e-01	1.48e-01	0.166
full, original bound	0e+00	3/6	9.41e-01	1.48e-01	0.161
fixed depth 4	0e+00	3/6	9.41e-01	1.48e-01	0.019
joint, original bound	1e-04	3/6	9.41e-01	1.48e-01	0.150
full, original bound	1e-04	3/6	9.41e-01	1.48e-01	0.149
fixed depth 4	1e-04	3/6	9.41e-01	1.48e-01	0.018

Original-bound joint recovery succeeds in 3/6 noiseless cases. The original joint and full rules retain all six probes and reach a full-space basis; this control supplies no compression benefit. The maximum parameter error is 0.94, and maximum held-out relative error 0.15. Independent projected gradients and Hessian diagnostics distinguish these poor fits from unconverged optimization, as summarized below. Small projected residuals at the incorrect boundary fits are compatible with constrained stationarity.

Table 8: Failed noiseless original-bound joint fits. The projected-gradient residual is $\|\theta - \Pi_{[-.5, .5]^3}(\theta - g)\|_2$ using an independent gradient. The smallest Hessian eigenvalue is estimated by central differences of independent gradients with step 10^{-5} , only at interior fits.

estimate	param. error	loss	projected residual	min. eigenvalue
(0.4169, 0.5000, -0.5000)	9.41e-01	7.16e-02	5.61e-10	—
(-0.1520, 0.3907, -0.1282)	1.51e-01	3.98e-06	1.74e-09	1.56e-03
(0.5000, -0.0159, -0.5000)	7.94e-01	6.79e-03	2.26e-11	—

An incorrect interior fit with positive numerical Hessian provides evidence of a nonglobal local minimum. Neither observation proves the global structure of the inverse problem. The observation Jacobian is formed independently at each truth. Its singular values are retained, and both Jacobians have rank three with condition numbers 52.8 and 41.1. This is local evidence only. Cosine is even, so A and $-A$ have identical responses whenever both are admissible. The fixed positive diagonal and parameter box exclude that specific sign reversal here; they do not prove global uniqueness. Successful runs also do not establish global optimizer convergence.

5.4. Larger learning problems with a feasible-direction test. To exercise active compression, use open chains of $n = 256$ and 1024 sites, with

$$A(\theta) = 0.7I - 0.22(1 + 0.3\theta_3)K + 0.18\theta_1 \text{diag}(w_{.25}) + 0.18\theta_2 \text{diag}(w_{.75}), \quad (w_c)_i = e^{-(i/n-c)^2/(2 \cdot .09^2)}. \quad (5.2)$$

The parameter box remains $[-0.5, 0.5]^3$. Twelve real preparations are the normalized columns

$$b_{c,k,a}(i) \propto e^{-\frac{1}{2}((i-nc)/(.035n)-a)^2} \cos(ki), \quad a \in \{-.12, -.04, .04, .12\}. \quad (5.3)$$

The three center/carrier pairs are $(c, k) = (.25, 0)$, $(.5, \pi/3)$, and $(.75, \pi/2)$. Carrier wave numbers help separate sensitivity to hopping and potential amplitudes. Lattice spacing, hopping, and time units stay fixed while envelope widths grow with the chain; this is not a continuum mesh-refinement experiment. The changed preparations, shorter times, and larger positive diagonal define a separate, pilot-informed measurement design. They are not an isolated ablation proving which change resolves the earlier recovery failures.

Training times are $(1, 2, 3)$ and held-out times $(1.5, 2.5, 4)$. Three truths, $(.22, -.18, .12)$, $(-.16, .24, -.14)$, and $(.08, .13, .23)$, use starts $(0, 0, 0)$ and $(-.3, .3, -.25)$. Each has a noiseless data set and two independent symmetric Gaussian amplitude-noise realizations of scale 10^{-4} . The complete factorial design crosses these settings with both sizes, four methods, and stationarity targets 10^{-5} and 10^{-7} . These remain simulated coherent-amplitude observations, not finite-shot measurements.

All four methods use projected damped Gauss–Newton steps from the available response derivatives, with a projected-gradient fallback. Joint and full certification use the refined bounds, tightening tolerance until Eq. (3.13) is negative and the evaluated gradient-error bound is at most one fifth of the projected-gradient norm. The stopping test adds that bound to the projected-gradient norm and compares the sum to the stationarity target. A full-rank depth-4 control and an independent augmented-reference-gradient control use the same step and line-search procedure without a gradient certificate. At most 35 accepted iterations and 20 backtracking trials per iteration are allowed.

Armijo acceptance uses independently computed sparse exponential actions of the original loss for every method. These are floating-point evaluations; we certify the feasible direction in exact arithmetic, not the complete line search. All acceptance costs, norm-estimation and transpose actions are counted. Final validation and a post hoc independent check of every computed gradient are timed and counted separately from optimization. Reference gradients do not influence the joint, full, or depth-4 directions. Stored trajectories distinguish evaluation points from accepted iterates.

Table 9: Larger learning study, with 18 runs per size, stationarity target, and method. Recovery uses parameter error below 10^{-3} . Rank and basis dimension are ranges across all gradient evaluations, including tolerance refinement; held-out error is the maximum. Time is the median full optimization time including independent loss acceptance, excluding post hoc audits.

n	stopping target	method	recovered	rank	basis dim.	held-out error	time (s)
256	1e-05	joint	18/18	6–9	84–144	4.4e-05	0.047
256	1e-05	full certified	18/18	12	168	4.4e-05	0.038
256	1e-05	fixed depth 4	18/18	12	48	4.4e-05	0.017
256	1e-05	reference gradient	18/18	12	–	4.4e-05	0.048
256	1e-07	joint	18/18	6–12	84–216	4.4e-05	0.076
256	1e-07	full certified	18/18	12	168	4.4e-05	0.051
256	1e-07	fixed depth 4	18/18	12	48	4.4e-05	0.022
256	1e-07	reference gradient	18/18	12	–	4.4e-05	0.063
1024	1e-05	joint	18/18	6–9	84–144	4.4e-05	0.082
1024	1e-05	full certified	18/18	12	168	4.4e-05	0.072
1024	1e-05	fixed depth 4	18/18	12	48	4.4e-05	0.037
1024	1e-05	reference gradient	18/18	12	–	4.4e-05	0.143
1024	1e-07	joint	18/18	6–12	84–216	4.3e-05	0.133
1024	1e-07	full certified	18/18	12	168	4.3e-05	0.096
1024	1e-07	fixed depth 4	18/18	12	48	4.3e-05	0.050
1024	1e-07	reference gradient	18/18	12	–	4.3e-05	0.190

The larger study contains 288 runs. Joint recovery meets parameter error below 10^{-3} in 72/72 cases. Across its gradient evaluations, basis dimension is 8.2–84.4 percent of the ambient dimension. The largest independent gradient discrepancy is 2.74×10^{-8} ; 0 evaluated bounds are exceeded by more than 10^{-12} . The additive allowance is a numerical audit threshold, not part of the exact-arithmetic

Table 10: Joint/full learning cost at each size and stopping target. Work entries are medians across the same 18 cases, in joint/full order. Total Hamiltonian work includes independent loss acceptance during optimization; post hoc audit work is excluded. The last column is joint/full median runtime.

n	stopping target	Krylov work	total Hamiltonian work	time ratio
256	1e-05	396/504	2352/2460	1.23
256	1e-07	792/672	3384/3288	1.50
1024	1e-05	396/504	2352/2460	1.13
1024	1e-07	792/672	3384/3288	1.39

certificate. At the looser stopping target, joint selection lowers the median Krylov work from 504 to 396, but the independent acceptance actions reduce the saving in total Hamiltonian work. At the stricter target, repeated refinement raises joint Krylov work to 792 versus 672 for full certification. Joint selection is slower than full certification in all four size/tolerance groups, and the depth-4 control remains faster. Active compression and successful recovery therefore establish feasibility, not a net speed advantage. All methods reach the requested independent projected-stationarity threshold. The reduced matrices occupy fewer than n columns throughout; basis storage is recorded, but peak process memory is not measured.

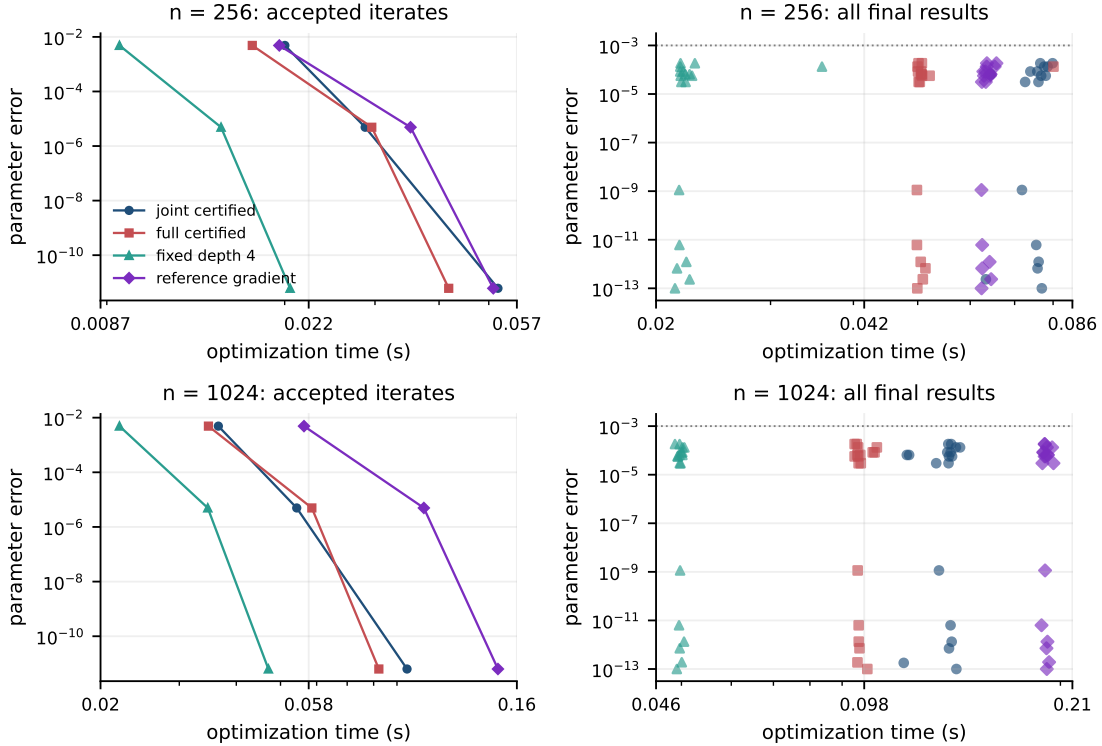


Fig. 5: Larger learning at stationarity target 10^{-7} . Left: accepted iterates for the first noiseless truth and zero start at each size. Right: all 18 final results per method, including both noise realizations and both starts. Optimization times include independent loss acceptance; audit time is excluded. Parameter error is evaluated against the generating truth.

6. Limitations and conclusion. The method provides an exact-arithmetic gradient bound and a finite-grid work optimum. Upward rounding of the tail evaluation does not certify row sums, SVD truncation, the computed basis, reduced eigensystems, or gradient assembly. The reported errors are discrepancies from an independently implemented floating-point reference. High-precision or interval end-to-end certification remains outside the present implementation.

The comparisons expose the main practical limitation: even the refined certificate can require substantially more depth than observed accuracy needs. A compressed certified computation must therefore be

judged against certified and uncertified alternatives separately. Spectrum-adapted bounds [8, 16, 17] and a cost model incorporating orthogonalization may further improve selection. The incremental and projected-adjoint controls strengthen the executable comparisons, but they are not an exhaustive state-of-the-art study. The tight-binding studies are simulated; their success or failure cannot be extrapolated to device Hamiltonian identification [22–24].

The resulting contribution is a reproducible study of the benefits and costs of certifying block matrix-function gradients. It supplies actual parameter recovery and independent references while preserving unfavorable cases. Practical usefulness depends on probe correlation, required accuracy, and whether a deterministic bound is itself an application requirement.

7. Data and code availability. Source, tests, pinned matrix checksums, complete numerical records, and figure scripts are in the accompanying repository at <https://github.com/thmolena/Block-Matrix-Products-for-Quantum-Operator-Learning>. Reproduction and validation commands are given in the repository README. The implementation SHA-256 digest begins 5ab68b10d8c8 and identifies the executable source and dependency specification; its full value is stored with the results. The timing-independent numerical digest begins f4a07060dcae. It detects changes in rounded numerical content, not scientific validity. External archives are downloaded, not redistributed. No archival DOI is claimed for this revision.

A. Positive tail evaluation. Let q be the first even integer greater than d , and $z = |t|\rho$. Successive even response terms have ratio at most $\zeta_0 = z^2/((q+1)(q+2))$; derivative terms have ratio at most $\zeta_1 = z^2/(q(q+1))$. If the relevant ratio is below one,

$$R_d^{(0)} \leq \frac{z^q}{q!(1-\zeta_0)}, \quad R_d^{(1)} \leq \frac{q|t|^q \rho^{q-1}}{q!(1-\zeta_1)}. \quad (\text{A.1})$$

Otherwise the implementation marks that tail infeasible and advances the depth. The formulas use 80-digit decimal arithmetic followed by an upward binary64 conversion. Response and derivative tails use their own degrees. The numerical implementation is not a full interval-arithmetic evaluation of the theorem. For the Chebyshev tails let $a = z/2$ and $b_q = 2a^q/q!$. Their successive-term ratios are bounded by $\xi_0 = a^2/((q+1)(q+2))$ and $\xi_1 = a^2(q+2)/(q^2(q+1))$, giving

$$E_d^{(0)} \leq \frac{b_q}{1-\xi_0}, \quad E_d^{(1)} \leq \frac{q^2 b_q}{\rho(1-\xi_1)}. \quad (\text{A.2})$$

An invalid denominator marks only that tail infeasible. Compared with the Taylor majorants, the first omitted response term shrinks by $2/2^q$, the derivative term by $2q/2^q \leq 1$, and both ratios decrease. Thus this tail refinement is never worse under the same enclosure and polynomial degree.

B. Independent references and checks. For p directions, assemble a sparse block lower-triangular matrix $\mathcal{A}$ with A on each of its $p+1$ diagonal blocks and D_j in block $(j+1, 1)$. Applying e^{-itA} to $(B, 0, \dots, 0)^T$ produces $e^{-itA}B$ in its first block and the exponential Fréchet actions in the remaining blocks [13, 21]. Their real parts give cosine responses and derivatives. SciPy’s exponential-action algorithm operates on this sparse augmented matrix; it does not use the projected cosine routine. The augmented problem has dimension $(p+1)n$ and is timed separately.

Tests compare clustered-eigenvalue divided differences and off-diagonal projected gradients with an independent dense exponential Fréchet routine. Explicit monomial sums check both exactness degrees and exhibit failure beyond the derivative degree. An exhaustive enumeration checks the discrete joint work minimum. The Si2–Si5H12 depth-28 reference discrepancies are additionally recorded against the augmented construction; depth agreement alone is not used as an independent validation argument.

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